Game Design Document: *Grimm Dominion*

(Working Title – A Dark Fairytale Asymmetric RTS for pc)

Overview

Genre & Platform: Asymmetric real-time strategy (RTS), 4-vs-4 multiplayer; PC.
High Concept: *Grimm Dominion* is a dark fairytale twist on the classic "MOBA" gameplay. Four players serve the all-powerful **Dark Lord** trapped in a gothic castle, each commanding hordes of AI monsters while ransacking the countryside looking for the Heroes of Light. Four opposing players are **Rebellious Heroes**, hiding and growing in power across a procedurally generated kingdom. All heroes are created each with unique playstyles and resource mechanics The tone is moody and gothic yet laced with wry humor, parodying fantasy tropes. Inspired by Warcraft III's custom game *Island Defense* (Island Defense v3.0.3g | HIVE) and modern 4v4 titles where multiple heroes face off, *Grimm Dominion* offers a fresh, mobile-friendly take on asymmetric strategy.

Key Features:

- *Asymmetric 4v4 Gameplay:* Four **Dark Heros** versus four **Light Heroes** with entirely different abilities, goals, and perspectives.

- *Unique Hero Mechanics:* Each hero has a distinct power-gathering loop (e.g. Goblin loots gold; Alchemist crafts potions) encouraging diverse team strategies.

- *AI-Controlled Minions:* The Dark Heroes summon **monstrous units** (scouts, tanks, priests) that roam with AI behaviors. Limited direct control means strategy is about clever unit deployment and prediction rather than micromanagement.

- *Stealth and Hunt:* Heroes start weaker and **hide** in a vast, shadowy kingdom. The Dark Heroes must **hunt** them using their Main Hero or use their units' independent vision and behavior patterns, creating a cat-and-mouse dynamic.

- *Procedurally Generated Maps:* Each match takes place in a new, procedurally generated kingdom – from haunted forests to cursed villages – enhancing replayability and requiring adaptive strategies.

- *Dark Fairytale Aesthetic:* A visual style combining **cartoony** character designs with **gothic, macabre** environments (think *Warcraft III*'s stylized darkness (Reforged graphics, art direction - General Discussion - Warcraft III: Reforged Forums)). The game's narrative and art parody classic fairytales with a twisted sense of humor

(Overlord (2007 video game) - Wikipedia).

- *Fair Free-to-Play:** Monetization is **cosmetics-only** – all gameplay content is unlocked via play. Cosmetics (skins, visual effects) let players personalize characters without affecting balance, following the proven model of games like *League of Legends* (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar).

Gameplay and Core Mechanics

Match Structure: Matches are **4v4 real-time battles** lasting ~40 minutes (target). The Dark Hero's goal is to eliminate all four heroes; the Heroes' goal is to kill the Dark Lord of the castle (by storming the innermost sanctum of the dark castle) **before time runs out or they're hunted down**. It's a team-based struggle of **survival vs domination**:

- **Early Game:** Light Heroes spawn dispersed across the map, weak and hidden among shadows. They gather resources quietly: gold, ingredients, mana, etc. Meanwhile, the four Dark Heroes start in the Dark Castle. They accumulate *Dark Energy* (from a shared pool) over time and send out minions to scout and harass. The villains may also individually venture out to claim nearby villages or corrupted shrines. Stealth is key: heroes avoid monster patrols, and Dark Heroes probe likely hideouts.

- **Mid Game:** As heroes level up through their unique loops, Dark Heroes unlock stronger units and base upgrades. For example, a Vampire Count might upgrade his crypt and train bloodthirsty thralls, while the Necromancer unearths more undead soldiers. Skirmishes erupt as hunting parties clash with roaming heroes. Heroes might bait monsters or split up; Dark Heroes must decide whether to use resources now for offense or save for a stronger defense later.

- **Late Game:** Surviving Light Heroes ideally converge on the Dark Castle for a final assault. By now each hero is powerful, so coordination is crucial. The Dark Heroes fortify their halls and amass a larger army of monsters. A climactic battle unfolds at the castle gate. Victory goes to the side that best leveraged early-game gains: if Dark Heroes weakened the intruders earlier, their defense might hold; if heroes remained at full power, the castle falls. This mirrors the original one-versus-many dynamic but scaled to four villains.

- **Core Mechanics:**

Core Mechanics:

- **Asymmetrical Roles:** Each Dark Hero functions like an RTS base commander as in the original design, but now with personal abilities. Villains manage a personal resource pool (e.g. *Souls, Blood, ecl*) to summon units or cast powerful spells. They issue orders to groups of monsters via a simple interface – for instance, selecting a patrol route or

attack target. Players don't micro-manage individual units; they deploy minions and direct them generally, with the unit's behavior taking over from there. Light Heroes play in third-person/action-RPG style, using cover and special abilities to slip past foes. The asymmetry means each player's experience is very different: Dark Heroes juggle base-building and macro strategy, while Light Heroes focus on exploration and personal upgrades.

- **Fog of War & Stealth:** The map is dark to Dark Heroes by default; only where their minions roam is revealed. Heroes are invisible unless a monster's vision cone spots them for a moment. This mechanic (and abilities like the Goblin's cloak) give heroes a chance to evade detection. Some villains have vision-enhancing minions (like bat swarms or scrying bats). Certain villains may have vulnerabilities to light or magic (e.g. Vampires burn in daylight), adding risk when they stray too far from cover.

- **The Dark Castle (Villain Base):** The Dark Castle is a large fortress partitioned into quarters for each Servant. It contains quarters for each dark hero, which enables them to upgrade or rest. Each Dark Hero returns here to spend collected resources, heal, and research new abilities. For example, the Necromancer might have a bone altar to sacrifice souls for upgrades, while the Demon Warlord uses a hellforge for demonic armaments. The castle itself can be upgraded by any villain (sharing fortified walls or traps) or selectively by individuals (each may spend resources on his own chamber). At the heart of the castle is the Dark Lord's antechamber, which is the goal of the heroes of light to destroy the dark lord slumbering within to regain his power.

- **Cooperation and Rivalry:** Without a single overlord, the four Dark Heroes navigate a tense partnership. They share a resource pool and common objectives (defend the castle, hunt heroes), but scarce resources and personal glory tempt betrayal. For instance, two villains might team up to ambush a hero, but fight each other over treasure. Dynamic alliances and backstabbing are possible – the social interplay adds depth. The game supports this by allowing limited diplomacy (e.g. gift an ally health or coin, or block access to a shared upgrade) so that strategies evolve on the fly.

- **Victory Conditions:**

    - *Dark Heros Wins:* if all four heroes are killed **before** the Dark Lord is slain. Heroes do **not** respawn once dead (though downed/bleeding-out states or revival items could be considered to avoid early eliminations – see **Systems**). Eliminated heroes might spectate or possibly assist as spirits (TBD in polish phase).

    - *Heroes Win:* if they kill the Dark Lord. This typically requires breaching the castle. The Dark Lord's castle acts as a final "boss area" – heroes may need to break a gate or disable wards to reach the Dark Lord. The Dark players will likely make a

last stand here, aided by any remaining minions or castle defenses.

- A **stalemate timer** (e.g. 40 minutes) may force a conclusion: if time expires, the Dark Lord's dark magic fully corrupts the land, counting as a win for Dark heroes (encouraging heroes to eventually attack, avoiding infinite hide-and-seek).

Team Dynamics: The four heroes are **cooperative** – they can communicate and coordinate tactics. While each hero has an independent progression, sharing information (like a monster's position or the discovered tavern location) is crucial. A lone hero caught by the Dark Lord's forces can be quickly overwhelmed, so teamwork (or at least timing their power spikes together) is encouraged for the endgame. The Dark heros, conversely, work alone with their AI minions. The design emphasizes **tension and mind-games**: heroes must decide when to stay hidden versus when to team up for objectives; the Dark Lord must anticipate hero behaviors and capitalize on any mistake (like a revealed position).

Roles and Characters

The game's asymmetry is embodied in its roles: one **Dark Lord** vs. four **Heroes**. The **Dark Lord** is an evil overlord character with strategic oversight and indirect control via minions. The **Heroes** are colorful, rebellious individuals with unique playstyles (each almost a "game within the game"). Below is a breakdown of these roles and their specific abilities, strengths, and weaknesses:

Dark Heroes (Servants of the Dark Lord)

Four malevolent overlords await their Dark Lord's return. Each commands their own minions and follow a unique power-progression path. All four share the Dark Castle as home base and draw from the same world resources, but they have distinct personalities, strengths, and thematic limitations:

Gameplay Role: The Dark Heros play like an RTS base commander / dungeon master: they gather resources and unleash AI-controlled **monster units** to do bidding, and spread corruption to infect the land. Once units are deployed, they behave according to type, but the Dark Heros can issue high-level commands or target priorities. They must manage **macro-strategy** (when to spawn units, where to send them) rather than micro every action. They also can employ limited **castle defenses** and possibly a few direct castle spells (if unlocked) as last resort. Importantly, the Dark Heros have **limited information** – they only know what minions can see or report.

Abilities & Resources:

- **(Resources):** The Dark Heros accumulate Resources by pillaging the land, These resources are used to **summon monster units** or activate special powers. If they kill a

hero, they gain a special resource that allow unique upgrades. They can also choose to **invest** energy into fortifying the castle (e.g. repairing gates or arming traps) versus spending on units to hunt.

- **Summoning Units:** The Dark Heros can spend energy to summon monsters. Summoning is not instant; each unit has a cast time and announces itself to the local area (giving attentive heroes a clue that a new threat is coming). Unit costs vary by strength – e.g. Scout: low cost, Tank: high cost. A **population cap** limits the total active units (for performance and balance), e.g. max 5 units at a time early, increasing to say 10 in late game. If cap is reached, the Dark Heros must wait for a unit to die or manually sacrifice one (perhaps via a "dismiss unit" command) to free up capacity.

- **Unit Command Interface:** Once units are spawned, the Dark Hero can select any unit (or group) via a minimap or unit list UI and issue simple orders: **Patrol** (an area or towards a waypoint), **Hunt** (follow tracks or head toward last sighting of a hero), **Rally/Defend** (return to castle or guard an area). After issuing an order, the unit's AI takes over to execute it. For example, ordering a Scout to "Hunt" near a forest village will send it there; it will then autonomously search that area, chasing any hero it finds. The Dark Hero retains **vision** through each unit – effectively acting as his eyes. He can also see a *small radius* around the castle by default (the "domain" vision).

- **Spells/Abilities:** To give the Dark Hero more active involvement, he may choose to focus on **a suite of abilities**: e.g. *"Scrying Orb"* (briefly reveals a chosen small area of the map for 5 seconds, at a high energy cost or long cooldown) or *"Fear Surge"* (a one-time shockwave from the castle that scares away nearby heroes or minions, used defensively), or *"Creeping Corruption"* (Taking direct control of a creep on the map for a limited time until its life runs out). These are unlocked via mid-match upgrades or tied to progression (so the Dark Lord isn't too powerful early). They provide clutch options but must be used sparingly.

- **Dark Lord Combat:** Each hero has a unique combat style (he might be a formidable spellcaster or melee boss). However, their power is enhanced by how much resource they save. Heroes will need to fight them like a boss battle, possibly while dealing with any remaining minions or traps.

Dark Lord's Monster Units: The Dark Lord commands several types of monster minions, each with distinct AI behavior patterns that heroes can learn and exploit. The core unit archetypes include:

Unit Type	Description & Role	AI Behavior (Logic)	Strengths	Weaknesses (Exploits)

Scout (e.g. “Imp” or “Gargoyle”)	Fast, small creature with high vision range but low HP and damage. Used to **explore** and reveal hiding heroes.	**Patrol AI:** Scouts wander along set paths or towards regions the Dark Lord flags. They investigate any sign of hero activity (footprints, open chests, etc.). If a scout spots a hero, it will pursue and *howl* to alert other monsters (pinging the hero’s location to the Dark Lord).	**Strength:** Very fast and observant; can quickly cover ground and chase fleeing heroes. Great for early detection.	**Weakness:** Frail – a single hero can kill a Scout if caught alone. Also has predictable patterns (e.g. always checks villages), so savvy heroes can avoid or ambush them.
Tank (e.g. “Golem” or “Ogre”)	Large, tough brute with high health and area-of-effect melee attacks. Used to **engage and soak** damage in fights.	**Aggressive AI:** Tanks head toward known hero locations or follow a general “sweep and clear” path. Once a hero is in range, a Tank will relentlessly chase and attack. They might also smash environmental objects (e.g. break barricaded doors that heroes hide behind).	**Strength:** Very durable; can absorb a lot of hero attacks. Dangerous up close, with heavy damage if it connects. Useful for sieging any fortifications.	**Weakness:** Slow movement and slow attack speed. Easy to kite or outrun. Can be led into traps or rough terrain that slows them further. If isolated, heroes can wear it down with ranged attacks.
Priest (e.g. “Dark Acolyte” or “Necromancer”)	Caster support unit with healing and detection	**Support AI:** Priests follow behind other	**Strength:** Keeps the monster army sustained	**Weakness:** Low direct combat ability – a Priest

	magic. Medium HP, weak offense. Used to **support other minions** and counter hero trickery.	monsters, preferring to stay at mid-range. They periodically **heal** nearby monsters or cast buffs (e.g. damage resistance). If a hero remains in line-of-sight for a moment, a Priest may cast a **Reveal spell** that temporarily outlines the hero (negating stealth for a few seconds). They may also avoid direct combat, retreating if no allies protect them.	– e.g. healing a Tank to prolong a fight. The Reveal ability can expose stealthy heroes or invisible tricks. Can dispel simple illusions or Alchemist's smoke bomb, etc.	alone is easy prey. Tends to follow predictable behavior (e.g. always heals the lowest-health monster), which heroes can exploit (bait the Priest into a vulnerable position). Removing Priests first is a common hero tactic to weaken the Dark Lord's forces.
(Advanced) *Special – Optional additional unit types to unlock via meta-progressio n: e.g.* ***Assassin*** *(fast melee damage dealer that can climb obstacles),* ***Beast*** *(a mid-tier balanced creature like a werewolf), etc.* These would provide more variety as the					

Dark Lord levels up account progress. For the core design, the above three cover the basic needs (scouting, combat, support).

The interplay of these units' behaviors creates depth. For example, a **Scout** might spot a Goblin hero briefly and howl, causing a nearby **Tank** to lumber that way. The Goblin could hide, letting the Scout run past, then slip out – but if a **Priest** arrives and casts Reveal, the Goblin might be exposed. Heroes can learn patterns (e.g. priests hang back) and plan ambushes (perhaps sniping a priest first). The Dark Lord, knowing these weaknesses, may send mixed groups (a Tank with a Priest support) or use scouts to bait heroes into the path of a Tank.

Dark Hero Strategy and Experience: Playing a Dark Lord feels like being the "director" of a horror movie – setting monsters loose and **reacting** to the heroes' actions. It's a mix of strategy and deduction: you have incomplete info and must predict where the heroes might be powering up. Did the Goblin likely head to the nearest village for gold? Is the Alchemist gathering in the swamp biome? A Dark Hero monitors his minions' output (if a Scout dies unexpectedly, perhaps it ran into a hero ambush; if no contact for a while, heroes might be ghosting around). The **wry humor** comes through the Dark Hero's commentary and the often inept but lovable AI minions (who might say "Yes, master!" or argue amongst themselves). Despite the humor, the Dark Hero player will feel a genuine sense of predatory tension – they are the hunter but also **outnumbered**, so any mistake or wasted resource could mean the heroes grow too strong. Good Dark Hero play involves **divide and conquer**: catching lone heroes, disrupting their resource gains, and not giving them time to all power up. Poor play (e.g. being too passive) might let the heroes collect gold and ingredients unopposed, eventually making them a nearly unstoppable force (Island Defense v3.0.3g | HIVE). Balancing this asymmetry is critical: the design must ensure the Dark Lord has a fair chance even if one hero is eliminated early (or conversely, that heroes can recover if one is picked off).

The Rebellious Heroes (4 Players)

The four Hero characters are distinct archetypes with unique gameplay loops and personality. They've all rebelled against the Dark Lord's tyranny and fled into the wild, each using different methods to grow strong enough for a revenge showdown. All heroes share some common traits: they each have a **basic attack** (melee or ranged depending on hero), a few **active abilities** (movement skills, attacks, or utility), and generally weaker stats than monsters at the very start. They rely on **hit-and-run tactics**, **hiding**, and **power acquisition**. Each hero has a **different resource mechanic** to gain strength: this asymmetry means each hero player's

experience is unique and encourages the team to cooperate (e.g. the Knight can fight monsters head-on to protect the Alchemist while she gathers ingredients).

Below is an outline of each hero, including their power sources and special mechanics:

1. The Goblin Outlaw – *Stealthy Thief*

Concept & Personality: A small, green Goblin who has defected from the Dark Lord's minions to join the heroes. He's cunning, greedy, and quick-witted. The Goblin brings a comedic flair – he cracks sarcastic jokes about the Dark Lord's "lousy pay for minions" and mutters to himself when stealing gold. Visually, he's hunched and sneaky, wearing a patchwork of stolen armor pieces.

Unique Gameplay Loop: Gold Acquisition & Upgrades. The Goblin gains power by accumulating **Gold** and spending it on upgrades to improve himself. He is essentially the "economy" hero, akin to a rogue/thief class.

- **Gold Sources:** Gold coins are dropped by slain monsters and can also be found by **stealing from villages** on the map. Small NPC villages (or cottages) exist in the kingdom; the Goblin can pickpocket these settlements or find stashes of coins in chests. However, raiding a village might attract attention – if a monster sees the Goblin in a village, it may infer a hero presence there.

- **Stealth Mechanic:** The Goblin is **invisible by default** unless an enemy has continuous line-of-sight on him for >2 seconds (as noted earlier). This makes him the hardest hero to detect. He can dart between shadows and break line-of-sight to reset detection. He also has an active **Cloak ability** (with a cooldown) to turn fully invisible for a short duration regardless of line-of-sight, useful for escapes or sneaking into the castle. If he attacks or interacts while cloaked, he becomes briefly visible, so timing is key.

- **Hidden Tavern (Upgrade Shop):** Scattered somewhere on the map is a **Secret Tavern** (a neutral hiding spot run by shady NPCs) where the Goblin can spend his gold on upgrades. The location of this tavern is randomized each match. The Goblin must **discover it** (it could be in a hollow tree, an abandoned inn, etc.). At the tavern, a shop UI allows purchase of things like improved dagger (damage up), boots (speed up), gadgets (smoke bombs, trap disarms), or even one-time powerful items. These upgrades make the Goblin a progressively more lethal and slippery foe. **Risk:** The Goblin must be careful not to **lead monsters to the tavern**. If the Dark Lord learns the tavern location (e.g. a Scout follows the Goblin there), he could camp it or destroy it (denying the Goblin further upgrades). Thus, the Goblin often takes long, convoluted paths to the tavern and might decoy in another direction to ensure it stays hidden. This mechanic creates intense tension for the Goblin player – every tavern visit is a risky but rewarding trip.

- **Playstyle:** Early on, the Goblin avoids combat unless it's to land a quick kill on a lone Scout for gold. He scouts out villages for loot, using his stealth to avoid detection. Mid-game, after a few upgrades, he can contribute more in fights – e.g. using a **backstab ability** (bonus damage when attacking from stealth) or deploying a purchased **smoke bomb** to allow the team to escape a fight. In a final confrontation, a fully geared Goblin might disable traps or flank the Dark Lord for critical damage. His low HP means he *cannot* go toe-to-toe with heavy monsters – he relies on trickery and timing. For the team, the Goblin often serves as the **information gatherer and disabler**: sneaking ahead, marking enemy positions to teammates, stealing resources the Dark Lord might use (maybe he can even steal a bit of the Dark Lord's gold from under his nose!), and initiating ambushes with a stun from behind.

Goblin Strengths & Weaknesses:

- *Strengths:* Best stealth in game; fast movement; can become powerful if he amasses gold (scaleable progression); good at initiating ambush or escaping danger.

- *Weaknesses:* Frail if caught; low starting combat ability; dependent on finding the tavern and getting gold (if monsters guard villages or kill him before spending gold, his potential is wasted); team relies on him for certain tasks, so if he's eliminated, they lose a lot of utility.

2. The Alchemist – *Explorer & Potion Master*

Concept & Personality: A quirky human (or gnome) Alchemist who was once the Dark Lord's royal potion-brewer but switched sides. She (or he) has an insatiable curiosity for ingredients and a slightly mad-scientist demeanor. Wears a ragged alchemical robe with bottles clinking at the belt. Frequently mutters about formula ratios and keeps a journal of discoveries. Her humor comes from outlandish reactions when a potion fails or a sarcastic remark like, "If it explodes, it's probably fine!"

Unique Gameplay Loop: Ingredient Harvesting & Potion Crafting. The Alchemist gains power by **collecting exotic ingredients** from the world and combining them into potions that grant various enhancements. This is a high-risk, high-reward loop with elements of discovery.

- **Exploration for Ingredients:** The map is divided into **unique biomes/areas** – e.g. a Poisonous Swamp, an Enchanted Forest, a Crystal Cavern, etc. Each biome has specific **unique creatures or plants**. The Alchemist's goal is to explore and gather ingredients from these. For example, the swamp might have **Bog Blooms** (plants) and **Giant Leeches** (creatures); the Alchemist can harvest from these (perhaps needing to kill a creature or find a plant and spend a few seconds gathering). Other heroes see these as ordinary parts of the environment, but for the Alchemist they are key resources. The distribution is randomized each game, so the Alchemist never quite knows what

she'll find where – encouraging genuine exploration under pressure.

- **Potion Crafting Mechanics:** The Alchemist has a **portable mixing kit** (or maybe needs a safe spot like a campfire) to combine ingredients. She can mix **two ingredients at a time** (perhaps later upgrades allow three). The results depend on the combination, and importantly, **combinations are randomized each match** within a set of possibilities. For instance, Bog Bloom + Leech might create a *"Swamp Tonic"* that grants poison resistance; in another match, it could yield a *"Leech Elixir"* that provides life-steal on attacks. The game will have a large recipe list (with logical themes, but randomized outcomes so the Alchemist player has to experiment). **Discovery** is a big element: the Alchemist may discover a powerful buff potion by trial, which permanently adds that recipe to her journal for the match. There could even be a meta-progression aspect of gradually learning a master recipe list across many matches.

- **Potion Effects:** Potions can do things like: **buff stats** (e.g. strength, speed, defense), grant **abilities** (e.g. temporary invisibility, shape-shift appearance), or **modify the hero's playstyle** (e.g. a potion that increases damage but slowly drains HP). Some potions may also affect allies or monsters (for example, a **stink bomb** potion could repel monsters from an area, or a **pheromone potion** might make monsters fight each other briefly). The Alchemist can choose to consume a potion herself or throw it like a grenade in some cases. Visually, potions could even alter her appearance (a nod to the prompt: e.g. drinking a troll strength potion makes her arms muscly and skin slightly green – cartoonishly reflecting the buff). These visual changes add humor and also telegraph to the Dark Lord what buffs she might have if his units see her.

- **Risk/Reward:** Crafting takes time and the outcome is uncertain until discovered. If the Alchemist brews the wrong combo, it could result in a **failed potion** (wasting ingredients, possibly causing a small explosion that deals damage or noise – oops!). This not only sets back her progress but could alert nearby monsters. Thus, the Alchemist might want to find a **safe hiding spot** (maybe a quiet corner of the map) to do her mixing. She might ask teammates (like the Knight) to stand guard while she crafts. Also, traveling to diverse biomes means she might venture far alone – a risky proposition if a monster spots her. Good communication is needed so others know her last location.

- **Playstyle:** Initially, the Alchemist avoids fights, focusing on gathering. She has modest combat ability (maybe a weak dagger or crossbow for self-defense). One early ability might be a **Flash Powder** – a blinding flash to stun a pursuing monster briefly, allowing her to flee. As she crafts potions, she becomes a wildcard: depending on what she creates, she could temporarily become a front-line fighter (after a strength potion), a swift scout (after a speed potion), or a supportive healer (if she discovers a healing draught she can share with teammates). This adaptability is powerful if used cleverly. By late game, a well-prepared Alchemist could have multiple buffs stacked – though likely with some trade-offs. For example, she might have super strength and resilience, helping the Knight tank the Dark Lord's hits, but maybe at the cost of a constant burn (if one potion

had a side effect). The Alchemist's versatility makes her a **jack-of-all-trades**, but only if she has time to prepare. She might also save a couple of *special potions* to use during the final castle assault – e.g. one that **melts a castle gate** or **nullifies a Dark Lord spell** for a short time.

- **Team Contribution:** The Alchemist can share certain potion effects with teammates. Perhaps she can brew tonics and drop them for others to pick up. This encourages her to coordinate ("I've got a spare defense potion for whoever is going to tank the next fight"). Her knowledge from exploration also helps map awareness – she might be the first to identify key locations (like "I found an ancient tree in the north – likely one of the Witch's shrines, and also a good ingredient spot").

Alchemist Strengths & Weaknesses:

- *Strengths:* Highly adaptable – can buff herself and allies in many ways; her power can **scale sharply** if she discovers strong potion combos; contributes to team utility (buffs, debuffs for monsters); her exploratory nature means the team gains map knowledge.

- *Weaknesses:* Slow start – needs time to gather ingredients and experiment; relatively weak in direct combat without potion buffs; unpredictable – if luck is bad or mistakes happen, she might not get the buffs she needs; splitting off to explore makes her vulnerable if not careful. Additionally, the Dark Lord might deduce her presence if, say, monsters keep dying to strange alchemical traps in one area – he could focus patrols there to catch her.

3. The Exiled Knight – *Brave Melee Warrior*

Concept & Personality: A once-honorable knight of the kingdom, exiled after the Dark Lord's rise to power. This Knight has a tarnished armor, a big sword, and a bigger sense of duty. He's the most "traditional" hero – think a Prince Charming or brave knight archetype, but with a rough edge (perhaps he's been living in the wild, cape torn, armor dented). Despite the dark surroundings, he maintains a courageous (if slightly naive) demeanor, often uttering dramatic heroic lines that come off as endearingly earnest or humorously over-the-top. For example, he might declare, "For the kingdom!" in a low voice when charging a monster, or quip "You shall not terrorize these lands any longer!" to a very confused ogre. His fairytale trope is the gallant hero, played with a bit of ironic charm.

Unique Gameplay Loop: Valor Quests & Leveling. The Exiled Knight gains power through **combat and heroic deeds**, essentially a more straightforward RPG leveling system enhanced by performing special **quests/objectives** on the map.

- **Valor (Experience) and Leveling:** As the Knight fights and defeats the Dark Lord's monsters, he gains **Valor** (experience points). Unlike other heroes who rely on external

resources, the Knight's growth is tied to combat proficiency. After accumulating enough Valor, the Knight **levels up**, which increases his base stats (health, damage, perhaps unlocks a new ability or improves an existing one). This makes him progressively stronger in direct fights. The Knight starts slightly stronger in combat than other heroes (to fulfill his role as initial defender), but he truly shines later if he can chain together victories.

- **Heroic Deeds (Map Objectives):** In addition to generic XP from kills, the Knight can undertake small **procedural quests** that appear on the map as opportunities, reflecting classic heroic deeds: for example, **"Rescue the Villagers"** – a village might be under attack by neutral creeps or minor monsters; if the Knight arrives and drives them off, the grateful villagers reward him with a Valor boost or a buff (and perhaps some info or an item). Another deed could be **"Retrieve the Sacred Artifact"** – a legendary sword stuck in a stone in some clearing, guarded by monsters. By claiming it, the Knight not only gains a better weapon (temporarily or permanently for that match) but also earns a chunk of Valor. These mini-objectives are optional but provide significant rewards. They are also broadcast subtly (heroes might get rumors or map markers for them). The Dark Lord might catch on and try to ambush these locations – creating hotspots of conflict.

- **Gear Upgrades:** While the Knight doesn't collect gold to shop, he can improve gear via found **relics or crafting**. For instance, monster loot might occasionally drop a **Rune** that the Knight can socket into his sword for +Attack, or a **piece of armor** from a defeated elite monster. Also, completing the above heroic deeds might grant gear: e.g. the Sacred Artifact quest literally gives him a stronger sword for the remainder of the match (or until he falls). This system ensures the Knight's progression isn't solely numeric leveling but also has tangible equipment upgrades. (In meta-progression, players could unlock different starting loadouts or cosmetic weapon skins for the Knight.)

- **Abilities:** The Knight's abilities are oriented around melee combat and protection: e.g. a **Shield Bash** that can stun a monster, a **Whirlwind Attack** hitting multiple foes around him, and possibly a **Defensive Stance** that raises his defenses for a short time (helpful when tanking for an ally). He might also have a minor **Healing Prayer** to slowly heal himself or an ally out of combat, reflecting his role as a guardian. These abilities improve or unlock as he levels up. For example, at level 3 he might unlock a short-range **Charge** ability to close distance.

- **Playstyle:** The Exiled Knight often acts as the team's **front-liner and distraction**. Early on, while others hide, the Knight is paradoxically the most likely to **engage** a monster. He might deliberately draw a Scout's attention, then lead it on a chase while the Goblin and Alchemist do their tasks elsewhere. Because of his higher initial toughness, he can survive situations others can't, though he must still be cautious — a Tank or too many enemies will overwhelm him. As he levels, small skirmishes become easier and he can even proactively clear out areas of minor enemies, making them safe for his teammates (e.g. clearing a wolf den so the Alchemist can gather wolf fangs peacefully). However,

fighting has consequences: noise and dead monsters can clue the Dark Lord into hero activity hotspots. So the Knight's player balances the benefit of Valor vs the risk of revealing himself. The Knight can also serve as a **bodyguard**: if, say, the Witch is channeling a shrine (see below), the Knight can stand guard and fend off any incoming threats, gaining Valor in the process. In the final battle, the Knight is usually the one **leading the charge** into the castle, and with enough valor-fueled levels, he can go toe-to-toe with the Dark Lord for a time – taking hits while the others exploit openings.

- **Team Contribution:** Morale and protection. The Knight might have an aura ability that boosts allies' defense when they're near him (reinforcing grouping). He can rescue allies who are chased by intercepting monsters. His quests often benefit all (e.g. saving villagers might reduce the Dark Lord's local influence, slightly slowing the Dark Lord's resource gain in that region, or grant a temporary buff to all heroes like "Blessing of the Villagers" = minor regen for a couple minutes). This provides an incentive for the team to help him with heroic deeds.

Knight Strengths & Weaknesses:

- *Strengths:* Strong duelist, especially as he levels; can protect teammates; straightforward progression that steadily increases power (no complex requirements aside from fighting); potentially game-changing rewards from heroic quests (which also add narrative flair).

- *Weaknesses:* His actions are the **least stealthy** – fighting and completing quests often creates noise, light, or other signals the Dark Lord can detect. If the Dark Lord focuses on the Knight early with multiple minions, the Knight could be overwhelmed (he's not invincible). Also if he's too cautious and doesn't fight, he won't level – so he must take some risks. His effectiveness is very gear/level dependent; a Knight who falls behind in XP will struggle to contribute later. Compared to other heroes, he has fewer "tricks" – mostly brute force and courage, which can be outmaneuvered by clever enemies if he's isolated.

4. The Forest Witch – *Arcane Saboteur*

Concept & Personality: A mysterious witch from the deep woods who despises the Dark Lord's corruption of nature. She has an otherworldly vibe – possibly a fairy or druidic sorceress. Visually, she might have deer antlers or a crow feather cloak, glowing eyes, etc., fitting the dark fairytale aesthetic. She speaks in riddles or archaic language, often mocking the Dark Lord's "cheap parlor tricks" compared to her ancient magic. As a "hero," she's morally grey – she's helping the rebellion largely to reclaim her forest, not out of altruism, which creates fun banter with the Knight's earnest heroism (e.g., Knight: "We must save the kingdom!" Witch: "I care only for my woods, knight. But if it means slaying that fool in the castle, so be it."). Her humor is dry and biting.

Unique Gameplay Loop: Arcane Energy & Rituals. The Witch gains power by clickping into the **magical ley lines** of the kingdom – performing rituals at ancient shrines to absorb arcane energy. In doing so, she also **weakens the Dark Lord's grip** on the land, acting as a natural counter to his evil magic. She's something of a support/sabotage hero with a magic growth system.

- **Ley Line Shrines:** The map contains a few (e.g. three) **Arcane Shrines** or totems scattered in different regions. These could be thematic – like an old World Tree, a Circle of Mushrooms, a Standing Stone with runes, etc. At the start, these shrines are dormant or corrupted by the Dark Lord's influence. The Witch can locate and travel to these shrines and perform a **Ritual Channel** (e.g. stand within a circle for 10 seconds uninterrupted). Doing so **purifies the shrine** and lets her absorb its energy. Each shrine channel gives the Witch a large chunk of **Arcane Power** (think of it as her XP) and often unlocks a new spell or augment for her. For example, the Forest Shrine might grant her a *Nature's Grasp* root spell, the Crypt Shrine might grant a *Spectral Form* ability, etc. This effectively means the Witch's abilities **scale up** as she claims shrines – she might start with only a basic curse spell, but by game end have a full suite of spells if she got all shrines.

- **Arcane Power Meter:** In addition to discrete shrine rewards, the Witch has an Arcane Power meter that fills from various magical actions: conducting rituals, collecting **arcane reagents** (like magic crystals or spirit essence dropped by certain magical creatures), and possibly passively when near a shrine. This meter might increase her base magic damage or mana pool. Once filled to certain thresholds, it could improve her stats or allow her to "talent pick" an upgrade (e.g. choose to make one of her spells stronger). This is analogous to leveling, but tied to magic gathering rather than combat.

- **Sabotage Effects:** Crucially, when the Witch purifies a shrine, it has a **debuffing effect on the Dark Lord**. For instance, each claimed shrine could slightly **reduce the Dark Lord's resource income rate** (his Evil Energy trickle slows by 10% per shrine) or disable a minor perk of his. If she manages to claim all shrines, perhaps the Dark Lord loses a significant advantage (like his vision radius from minions is reduced or his castle's protective aura weakens). This directly helps the heroes and pressures the Dark Lord to contest the shrines. The Dark Lord will be alerted when a shrine is being challenged (e.g. "A surge of energy from the west – the Forest Shrine is under attack!"), likely prompting him to send forces there. This creates a dynamic objective on the map much like the Knight's quests, but with direct impact on the villain.

- **Spells and Abilities:** The Witch starts with a basic offensive spell (e.g. *Shadow Bolt*, a ranged magic attack) and a utility spell (*Hedgewalk* – a short blink/teleport or transforming into a swarm of crows to move unseen a short distance). As she gains power, she can unlock more: possibilities include *Entangling Vines* (rooting enemies in area), *Illusory Doppelganger* (create a fake copy of a hero to distract monsters), *Curse* (weaken a monster's attack or slowly drain its health), or **polymorph** fun (maybe turn a

small monster into a harmless animal briefly). One ultimate ability she might gain late is *"Fury of the Forest"* – summoning thorny roots or allied treants to fight for a short duration, turning the tables by adding her own AI minions in a fight. Her abilities trend toward **battlefield control and debuff** rather than raw damage (so she complements the Knight and others).

- **Playstyle:** The Witch must balance going after shrines with supporting her team. Early on, she may team up with the Knight because she's not very durable if caught alone. With one shrine claimed, she becomes noticeably stronger and gains more independence. She might prioritize claiming at least two shrines, then use her enhanced magic to join the others in fights or castle assault. If the Dark Lord heavily guards a remaining shrine, she can choose to skip it and stick with the team to fight directly – though they'll miss the debuff on the Dark Lord. Her presence in battles is subtle but powerful: e.g. she might cast **Glamour** on the Goblin to mask him from a monster's sight, or place a **hex** on a Tank making it deal less damage. Unlike the Knight, she avoids direct melee and stays at range or in cover, using hit-and-run spells. She has to be mindful of **Priest units**, as their Reveal spell can counter some of her trickery or remove her summoned illusions. In the final battle, a fully empowered Witch could nullify some of the Dark Lord's advantages – for example, if she had a spell to temporarily disable the Dark Lord's ability to summon new minions during the fight, that could be decisive.

- **Team Contribution:** The Witch is the **crowd-control and magical utility** specialist. She can coordinate with the Alchemist for interesting combos (e.g. Witch roots a group of enemies, Alchemist throws a potion AoE at them). She also provides strategic map control by removing the Dark Lord's shrine benefits. Teammates will often help her secure shrines because it benefits everyone. Communication is key: she might say "Cover me while I channel!" and the others fend off incoming monsters for those crucial seconds. Post-ritual, she can possibly bless the area – maybe shrines, once purified, become **safe zones** where heroes can heal or hide briefly (since Dark Lord's corruption is gone there). This gives heroes fallback points on the map.

Witch Strengths & Weaknesses:

- *Strengths:* Potent control spells and debuffs; scales in power similarly to the Dark Lord (she's basically drawing from the same source of power he uses, turning it against him); can significantly weaken the Dark Lord's macro-game via shrine debuffs; versatile magical toolkit by endgame; provides team support that no other hero can (disabling monsters, illusions, etc.).

- *Weaknesses:* Very vulnerable if caught early or alone (low HP/physical defense); her progression requires being at specific locations (shrines) which can be predicted and contested; slow casting times on rituals (if she's interrupted, she has to start over); requires game sense to use spells effectively (higher skill cap perhaps). If the Dark Lord

ignores other heroes and targets her shrines heavily, she might struggle to power up, relying on team to help push those objectives.

Heroes Summary Table: (Quick comparison of the four heroes)

Hero	Power Source	Mechanics	Strengths	Weaknesses
Goblin Outlaw	Gold (Loot from monsters & villages)	Stealth; Hidden Tavern for upgrades	- Nearly invisible when careful (stealth)- Fast and can scout safely- Powerful item upgrades if rich- Trickster tools (smoke bombs, traps)	- Lowest health; very squishy in combat- Needs tavern; vulnerable when spending gold- Low damage until upgraded- If caught early, can't contribute much
Alchemist	Ingredients (from creatures/biomes)	Exploration; Potion crafting (randomized recipes)	- Highly versatile via potion effects- Can buff entire team or debuff enemies- Strong scaling if many recipes found- Good map knowledge through exploration	- Weak early combat; relies on finding ingredients- Crafting can backfire (wasted time or self-harm)- Randomness means not guaranteed optimal buffs each game- Multi-tasking crafting while avoiding danger is challenging
Exiled Knight	Valor XP (from combat & quests)	Direct combat; RPG leveling; Heroic mini-quests	- Strong melee fighter, good for engaging enemies- Can protect allies and draw aggro- Steady,	- Loud and noticeable to the enemy (low stealth)- Progress tied to fighting – risky in asym game- If

			predictable progression- Quests can grant powerful rewards (gear, buffs)	unable to fight (hiding too much), he falls behind- Primarily close-range, can be kited by monsters without support
Forest Witch	Arcane Energy (from shrines & magic nodes)	Ritual channeling; Spellcasting & debuffs	- Excellent crowd control and support magic- Weakens Dark Lord's overall power via shrines- By late game, has diverse spells (illusion, damage, heal etc.)- Adds strategic objectives that benefit team	- Fragile if confronted head-on, especially early- Must channel at shrines (high risk of interruption)- Limited offense alone; relies on spells rather than raw damage- Targeted counter: enemy Priests can dispel or reveal her tricks

Each hero's unique loop means that over a match, the **team of heroes** ideally covers each other's weaknesses. For example, while the Knight and Goblin handle a monster threat, the Alchemist and Witch can focus on gathering resources in relative safety. Or the Goblin's intel can help the Witch avoid patrols on the way to a shrine. A coordinated team that leverages all four roles will have the best chance at overthrowing the Dark Lord. In contrast, the Dark Lord's advantage is unity of command – one will versus four separate agendas. A poorly coordinated hero team might get picked off one by one by the Dark Lord's focused strategy. It's this tension that makes *Grimm Dominion* compelling: asymmetric gameplay where knowledge, deception, and strategy are as important as quick reflexes.

Systems and Mechanics

This section details the underlying game systems that support the gameplay, including hidden mechanics like stealth visibility rules, procedural generation of the map, progression systems, and AI behavior subtleties. Ensuring these systems work harmoniously is key to a balanced and fun experience.

Stealth & Visibility System:

Stealth is a core pillar of the heroes' gameplay. The system works on line-of-sight and duration to prevent unfair sudden reveals while giving the Dark Lord chances to detect players who linger carelessly. Key rules:

- **Line-of-Sight (LoS):** All Dark Lord units (and the Dark Lord's castle) have a vision range with a frontal cone (for scouts) or 360° (for stationary sources, e.g. castle towers). Heroes outside this range or behind obstacles (walls, dense treeline) are **invisible**. There is a Fog of War covering the map for the Dark Lord – areas with no units have no vision at all. Heroes see the entire map (shrouded in style but not in terms of fog for them, as they need situational awareness in a different way), but they don't know where enemies are unless spotted.

- **Detection Delay:** When a hero enters a monster's line-of-sight, they aren't instantly revealed. A **"detection meter"** begins filling (taking ~2 seconds as noted for Goblin's ability). Each hero has a different detection threshold (Goblin perhaps 3 seconds base because of his innate stealthiness, Knight only 1.5 second due to clunky armor?). If the hero breaks LoS before the timer fills, the monster loses track. If the timer fills, that hero's **icon and position are pinged** to the Dark Lord (and the monster will actively pursue). The Dark Lord's client might get a subtle sound or visual ping "Hero spotted!" with an icon on the map that fades after a few seconds (if the hero goes out of view again). This gives the Dark Lord partial info – e.g. "a hero was seen near the river". It's then up to him to rally units to search that area.

- **Noise & Alerts:** Beyond vision, **sound cues** matter. If a hero fights or runs noisily (e.g. Knight clanking in armor, Goblin breaking into a chest with a loud creak, Alchemist's failed potion exploding), nearby monsters might investigate even without direct sight. We simulate this by having actions emit a "noise radius". In practice, if a hero uses a gunpowder-based bomb, all monsters in X range get an alert and may head toward that location. This prevents heroes from doing anything with impunity just because no monster was watching. It also encourages use of quieter methods (melee kills are quieter than explosives, etc.).

- **Camouflage & Disguises:** Some heroes have special stealth tricks: e.g. the Goblin's cloak ability, or perhaps a potion that lets the Alchemist **mask her scent** from monsters, or the Witch casting an illusion to appear as a neutral critter (temporary disguise). These can completely negate detection for a time, but have limits (duration or break on action). Monsters might ignore a disguised hero unless they do something obvious.

- **Monster Perception:** Different monsters have different perception. Scouts have long vision and quick detection (shorter delay), making them good at catching glimpses of fast-moving heroes. Tanks have narrower vision (they might even be somewhat dim-witted, only noticing heroes very close). Priests have moderate vision but can also

magically sense if a hero uses magic nearby (for example, the Witch casting a spell might ping a Priest due to magical disturbance). These differences mean heroes must adapt their stealth approach to the enemy present.

- **Lighting & Time (Optional):** For atmosphere, we might simulate day-night cycles or lighting conditions. At night, heroes might be a bit harder to spot at range, but perhaps the Dark Lord's minions use torches or have darkvision. Given mobile real-time constraints, a full day-night cycle might be too much, but we could have certain areas that are dark (inside caves, dense forest) where detection range is naturally shorter, and open well-lit areas (village with torches at night, etc.) where it's longer. The Witch could possibly extinguish lights magically to create darkness as a tactic.

In summary, the stealth system is designed so that **careful movement and terrain use** can let heroes operate under the Dark Lord's nose, but obvious actions or staying too long in one place can give them away. Heroes will feel tension as a scout wanders close – crouching behind a rock hoping it leaves. The Dark Lord, on the other hand, gets partial info and must interpret it – was that quick sighting by a Scout the Goblin or just a critter? He may send a Tank to investigate heavily, which could waste time if it was a false alarm. This asymmetry of information is crucial for suspense.

Procedural Map Generation:

Each match's kingdom map is procedurally generated to enhance replayability and prevent memorization of hiding spots. The map is a roughly circular region with the Dark Lord's castle at the center. Procedural generation works as follows:

- **Biome Layout:** We have several biome types (forest, swamp, hills, graveyard, village, etc.). The generator will create a patchwork of these biomes in the map, ensuring key ones appear (for variety) and spacing them out logically (e.g. swamp often near a river or lowland, villages on edges of forest, etc.). Each biome has distinct visuals and also **gameplay elements**: e.g., swamp slows movement but has rare ingredients, forests provide lots of hiding bush cover, open plains allow fast travel but little cover.

- **Key Landmarks:** Certain structures or landmarks are guaranteed but with randomized locations:

    - The **Secret Tavern** (for Goblin) – it will spawn in a somewhat secluded spot, possibly within a village or at a crossroads. Only when the Goblin gets close will he maybe get a hint (like a marker or an NPC whisper "psst, looking to buy something?").

    - The **Arcane Shrines** (for Witch) – spawn at three distinct biomes (e.g. one in forest, one in an old graveyard, one in mountain ruins) to force travel.

    - The **Hero Spawn Points** – the four heroes spawn in semi-random locations at map start, but likely near resources suited to them (Alchemist near a biome with some ingredients, Knight maybe near a small monster camp he can handle, Goblin near a village outskirts, Witch near a magical glade). This is to give each a fair starting chance to do their thing.

    - **Villages/NPC areas:** A few small villages, farmsteads or camps dot the map. These serve as gold sources for Goblin, quest locales for Knight (e.g. villagers to save), and just ambient world. Their positions move each game. They may also contain neutral NPCs (villagers, animals) that can be interacted with (Knight talks to them for a minor buff, Goblin steals, Dark Lord's monsters might attack them if passing through which could incidentally reveal monster positions to heroes via sound).

    - **Neutral Monsters/Creatures:** Besides the Dark Lord's units, the map might have neutral hazards – e.g. packs of wolves, giant spiders in a cave, etc. These present minor obstacles for heroes (and sources of XP or ingredients). The Knight's quests often involve these. Dark Lord units might ignore neutrals or kill them as they wander. Sometimes neutrals can even help heroes unintentionally (a bear might attack a Dark Lord's scout that gets too close). The neutral monster distribution is randomized.

- **Terrain & Pathing:** The generator ensures multiple pathways between areas, avoiding dead-ends where a hero could be trapped unfairly. For example, the castle is central, but heroes should have at least 3 or 4 different approach paths (so Dark Lord can't just bottle them in easily). Rivers, bridges, and maybe underground tunnels (one-way shortcuts) can appear. All terrain is made to be navigable by both heroes and AI (with slower movement in some terrain but not impassable except perhaps the castle walls which have gates). Some terrain features can be destroyed or altered (Knight could chop down a tree to block a path behind him, Dark Lord's Tank could smash through a breakable wall rather than going around). These are predefined in the map chunks and appear randomly.

- **Castle Design:** The Dark Lord's castle is semi-procedural but more static in layout. It might have outer walls and an inner keep. Gates that can be broken, perhaps a moat or one secret entrance (randomly one of a few possibilities – e.g. a sewer tunnel one game, a hidden postern door another) that observant heroes could find for a sneaky entry. The castle area also contains a **resource node** for the Dark Lord: maybe a "Well of Souls" that generates his Evil Energy. It's mainly flavor, but the Witch could target it with spells if she gets close, or Goblin could try to steal a bit. Destroying it might be an alternate win condition (or just disables Dark Lord's income). The castle also likely houses a couple of **static defenses** like arrow towers or cursed gargoyles that act as sentries (small AI that just guard the walls). These make a frontal assault costly without preparation (heroes

may want to take them out covertly or with ranged attacks).

- **Random Events:** To keep things dynamic, we can have a small pool of random events or features: e.g., a travelling merchant (neutral NPC) wandering through (Goblin might rob them, Knight might escort them for a reward), or a “Blood Moon” event at mid-game where the Dark Lord’s units get buffed for 60 seconds but also neutrals become aggressive to everyone. These should be used sparingly to add spice and not overshadow core gameplay, but they add to that fairytale unpredictability.

The procedural system will be carefully tuned so maps feel **familiar but fresh**. Landmarks prevent it from being completely random noise – players will learn patterns like “Usually there’s at least one village in the east or south” or “Shrines tend to be in distinct biome corners”. But the exact layout and distances change, which means the Dark Lord can’t just memorize “tavern always here” – he has to actually search, and heroes can’t always run to the same hiding spot each match. This unpredictability is important to avoid stale meta strategies and, combined with the variable hero power mechanics, should make each match story uniquely different.

AI Behavior & Exploitation:

We’ve outlined monster AI in the roles, but here we detail how it’s implemented and how players can interact with it. The AI is designed to be **competent but intentionally exploitable**, to reward skill and clever tactics (especially from the heroes’ side).

- **Behavior Trees:** Each monster type has a behavior tree dictating priorities. For example, **Scout AI**: Patrol -> if hears noise, investigate -> if sees hero, chase and call for help -> if hurt beyond X, maybe flee (or keep fighting if fearless imp?). **Tank AI**: Roam towards target -> smash obstacles -> attack hero if in range -> if no hero in sight, maybe idle or return. **Priest AI**: Follow strongest ally -> heal allies at <50% HP -> cast reveal on sighted hero -> avoid direct combat unless cornered. These behaviors will be scripted and can have slight randomization (maybe a Scout sometimes doubles back on its patrol unpredictably, etc.).

- **Alert System:** When one monster spots a hero or takes damage, nearby allied monsters get alerted and may shift their behavior (a bit like an aggro system). E.g., a Scout howling might flag all monsters within a certain radius or all idle monsters to move toward that scout’s location. This creates a **snowball effect** if heroes trigger an alarm: one slip-up can bring a lot of trouble. But it also means if heroes manage to silently eliminate a monster without it alerting (e.g. burst down a Scout before it howls), they avoid drawing others.

- **Exploiting AI:** Each monster’s weakness can be exploited by players who observe them:

    - Scouts might be drawn to any movement, so a Goblin could throw a rock (decoy item) to make a Scout run the wrong way. Or lead it into an area with a neutral monster that will attack it (e.g. kite a Scout near a sleeping ogre).

    - Tanks are slow and follow linearly; heroes can lead a Tank on a wild goose chase around rocks while others go do objectives. Tanks also might not fit through narrow paths, so heroes could escape where the tank has to take a long way around.

    - Priests, as noted, will try to stay with others – so if heroes manage to snipe the frontline monster, the priest might panic, making it easy to kill after. Also Priests might focus so much on healing that they don't run away while casting, so heroes can capitalize on that tunnel vision.

    - The AI doesn't cheat with full map knowledge; it reacts to stimuli. So a clever hero team that stays completely silent and unseen can technically have periods of doing whatever without any monster interference – but the game design ensures that to win (killing Dark Lord) they eventually must engage.

    - The Dark Lord can override some AI by ordering units, but once out of his immediate command, they still follow their nature. A common master strategy might be the Dark Lord sending a mixed group – heroes then have to pick which AI to exploit first (usually take out Priest, kite Tank, etc. as earlier discussed).

- **Difficulty Scaling:** Over time or based on Dark Lord actions, the AI might **scale up** in difficulty. For example, after 10 minutes, all monsters gain +10% movement speed (representing the Dark Lord's growing anger fueling them). Or if only one hero remains alive (lone survivor scenario), monsters become a bit more relentless to prevent a tedious hide-and-seek. These ensure the game reaches a climax.

- **Hero AI (if needed):** In case of players dropping or for potential single-player vs AI modes, we might have rudimentary AI for heroes, but primary scope is PvP so we won't detail that here (focus is on monster AI).

Combat & Abilities System:

- **Controls & Targeting:** Combat is streamlined for mobile. Heroes auto-attack with their basic attack when an enemy is in range (with an option for manual attack via a button for precise control). Abilities are triggered via buttons (with cooldown timers shown). Targeting for abilities can be done either directionally (swipe or click a target) or self-cast if buff. We ensure abilities are few and impactful rather than spammy.

- **Stats:** Under the hood, characters have stats like Health, Attack Damage, Armor (damage reduction), maybe Magic Power (for witches/alchemist's potion effects). The Knight and Tank have high health/armor, Goblin and Scout low, etc. Damage is calculated in a simple way to keep things understandable (e.g., subtract armor from damage or a % reduction formula). Heroes can see their HP and that of allies; the Dark Lord sees HP of his units (and maybe approximate health of heroes if his units spot them, but perhaps not exact).

- **Abilities & Cooldowns:** Each hero has ~3-4 abilities plus possibly an ultimate or item actives. Each ability has a cooldown (ranging few seconds for minor skills to a minute for powerful ultimates). The Dark Lord's units mostly just auto-attack and use innate skills per AI triggers (e.g. Priest auto-heal cooldown, Scout howl as ability with cooldown). The Dark Lord's own possible spells (like Scrying Orb) also have long cooldowns so they are periodic.

- **Damage Types & Resistances:** To add depth, there could be elemental or physical damage types. E.g. Alchemist potions might do fire damage (Tank might be resistant to physical but weak to fire?), Witch's curse might bypass armor. These can make certain heroes more effective against certain units. But we'll be cautious with complexity given mobile audience. At least a concept of "armor vs magic" might exist.

- **Friendly Fire:** Generally off for heroes (they can't damage each other with attacks – important for team trust). But some things like an Alchemist explosive could hurt allies if used recklessly (light damage) – this is more for humor than serious damage. Dark Lord's monsters can hurt neutrals and vice versa, but usually not each other (except if an Alchemist potion or Witch spell causes betrayal effects).

- **Healing & Death:** Heroes can recover health slowly over time when out of combat (regeneration) or faster via certain potions (Alchemist can brew healing) or Knight's minor heal ability. There might be limited consumables found (like one-time use healing salves in the world). Death is usually final in a match (no respawn). However, we could allow a *"downed state"* when a hero hits 0 HP – they crawl and can be revived by a teammate within, say, 30 seconds. This adds a dramatic rescue dynamic and is often used in team games to avoid players sitting out too early. The Dark Lord's units can execute a downed hero to finish them. This would encourage heroes to stick somewhat together to save each other. This detail would be refined in testing for balance. The Dark Lord's units themselves simply die when HP is depleted (heroes might target Priests to stop them resurrecting units if we had such an ability, but currently priests only heal, not resurrect, to keep Dark Lord from having too much recovery).

- **Castle Assault Endgame:** Combat at the castle might introduce area hazards: e.g. the Dark Lord could have a floor trap that spews fire (heroes must avoid), etc. The environment itself becomes part of the combat. The Dark Lord enemy (as a boss) might have multiple phases (like shield until minions are killed, then vulnerable). This is more in

boss design territory, but as a system note, the final fight should be the most intense combat-wise, leveraging all abilities heroes and Dark Lord have left.

- **Heroes Dying** - If a Hero Dies, they are resurrected as a Sidekick for another alive player with unique gameplay and limited capacity, but able to help their fellow player become stronger by gathering resources, helping with quests, or fighting.

Progression & Meta-Progression:

Beyond the match-to-match progression (heroes leveling in a match, Dark Lord accumulating resources in a match), *Grimm Dominion* features long-term progression to keep players engaged over many sessions. This includes unlocking new abilities, cosmetic rewards, and possibly new characters over time, but **without** introducing pay-to-win or unbalanced advantages.

- **Hero Progression (Meta):** Each hero character has a progression track (account-bound). By playing matches and earning experience (regardless of win or lose, though winning might grant bonus), players **level up that hero**. Leveling a hero could unlock:
    - **New Ability Variants:** For example, at account level 3 with the Goblin, you unlock an alternative upgrade in the tavern: “Poisoned Dagger” upgrade option (adds DoT damage on backstab) which you can choose in future matches instead of the default. Or the Alchemist might unlock new ingredient combinations (wider potion recipe pool) as she levels up persistently, meaning experienced Alchemist players have more possible potions (but importantly, not strictly more powerful, just more variety/options).
    - **Skill Tree / Perks:** We might have a light skill tree for each hero that offers minor tweaks. E.g. Knight can choose a perk at account level 5 that either gives an alternate starting weapon or starting armor to prioritize tankiness vs damage. These should be small enough not to upset balance too much, but allow personalization. Alternatively, perks could be selected pre-match (loadout style) once unlocked, giving some strategy (like Dead by Daylight’s perk system). We’ll ensure new players are not outclassed simply due to fewer perks – matchmaking can account for progression level, and perks will be sidegrades mostly.
    - **Unlocking Other Heroes:** If additional hero characters (beyond the initial four) are planned, players might unlock them via meta progression. For example, after enough play, a new hero class “The Ranger” or something could become available. However, since we have four core roles, any new hero would likely fit into similar asymmetry (could be DLC or seasonal additions). All such unlocks would be free (or earned) given monetization is cosmetics-only.

    - **Cosmetics & Titles:** Progression will also reward cosmetics (discussed more in Monetization). Players might earn a free skin at certain levels or a title like "Goblin King" after winning X matches as Goblin, etc., to show off.

- **Dark Hero Progression (Meta):** A Dark Heros player similarly has an account progression. This could unlock:

    - **New Monster Types:** e.g. at level 5, unlock **Assassin Imp** unit as a new summon option (fast melee glass cannon). Or at level 10, unlock **Lich** as an alternative to Priest (trades healing for offensive spells). These add variety to the Dark Heroes toolkit. We have to ensure that these new units are balanced – ideally not strictly stronger, just different. This lets advanced players customize their "loadout" of what 3 unit types to bring into a match. Before match, Dark Hero can choose, for example, between Priest or Lich as their support unit type, if both unlocked.

    - **Castle Upgrades:** Perhaps persistent upgrades to the castle or Dark Castle's personal abilities. For instance, a talent that increases starting Evil Energy or one that makes castle arrow towers stronger. Again, kept small to not make new players hopeless against the leveled Dark Heroes, but enough to feel progress. We might matchmake Dark Heroes by level too. Alternatively, instead of permanent stat boosts, these could just unlock more *options* (like traps you can place in castle, each match you can pick 2 trap types out of unlocked ones). This keeps things fair by giving variety rather than pure advantage.

    - **Cosmetics:** Skins for the Heroes (different appearances), skins for units (make your summoned monsters look like different creatures – purely visual), castle cosmetics, etc. Many of these could be tied to progression or events as free rewards or purchasable as well.

- **Account Progression & Rewards:** There will also be a general player account level that goes up as you play any role, granting basic rewards (currency for cosmetics, loot boxes for cosmetic items if we have those, etc.). Meta-missions like "Win 5 games as a Hero" or "Deal 1000 damage with monsters" encourage trying both sides. This keeps players engaged with all aspects of the game.

- **Balancing Progression:** We will carefully ensure that meta-progression does not significantly imbalance the core competitive gameplay. One method is to keep match-made games within similar progression tiers. Another is to focus unlocks on **sidegrades and cosmetics**. For example, a new unit type for Dark Lord isn't necessarily better, just a different style (maybe a flying unit that can hop walls but has low HP). A new hero ability variant might be situationally better but not an outright upgrade. The goal is that a skilled new player can still outplay a less-skilled veteran even with fewer unlocks, especially since game knowledge counts for a lot in asymmetrical

games. This approach is consistent with successful competitive titles that monetize via cosmetics and light progression without pay-to-win (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar).

User Interface (UI) & User Experience (UX): *(Note: UI is given its own main section below as requested, but we mention certain systems UI elements here as they relate to mechanics, such as HUD indicators, etc.)*

Key system UI features include:

- **Minimap:** All Heroes have a minimap. The Dark Heros minimap shows known terrain (from unit scouting) and pings when units alert. Heroes' minimap might show basic terrain and known landmarks (updated as they explore or share info). The minimap is crucial for spatial awareness on mobile's small screen.

- **Ping/Communication System:** Since typing is hard on mobile in real-time, a ping wheel allows heroes to mark things ("Go here," "Enemy spotted," "Need help"). A Dark Hero can also ping/flag areas for his units via the same mechanism (that's essentially how he gives move orders).

- **Inventory UI:** Goblin's gold count, Alchemist's ingredient pouch (with icons of collected items), Knight's maybe quest log indicator (showing an active quest like "Protect Village [0/1]" for a small XP bonus), Witch's arcane power meter and shrine status (like icons for each shrine that light up when cleansed) – these are all shown on the HUD in a minimal way. We use small icons on the side or top for these hero-specific resources. If an Alchemist picks up an ingredient, a small toast "Picked up *Red Mushroom*" shows and her ingredient UI updates (2/5 mushrooms for a potion, etc.).

- **Notifications:** When a hero gains a level or new ability (Knight leveling, Witch claiming shrine), a notification appears. When a hero is killed, everyone sees "[Hero Name] has been slain!" When a shrine is cleansed, Dark Lord sees " broke the curse on the Ancient Stone – your power wanes!". These messages add drama and convey important info.

The systems above all feed into giving players feedback and strategic info so they can make decisions. We will playtest extensively to tune parameters (detection times, resource gains, monster stat ratios) so that matches feel balanced – e.g. we don't want every game to always favor heroes or Dark Lord purely by system output. We'll use analytics from alpha tests to see win rates. As noted in design references, asymmetrical games require careful tuning so one side doesn't snowball too easily (Island Defense v3.0.3g | HIVE). Our design includes comeback mechanics (e.g. Dark Lord gets a boost if heroes take too long; heroes have teamwork synergy to overcome early losses) to keep matches dynamic.

User Interface (UI) & Controls

Designing an intuitive UI is critical, especially given the complexity of asymmetric gameplay and the constraints of a mobile touchscreen. We strive for a clean, diegetic interface that presents necessary info without overwhelming the small screen. Below, we outline the UI and controls for all Hero players, as well as general UI elements.

General Screen Layout:

- The game uses a **camera** perspective appropriate for each role: an isometric/top-down view for heroes that can be moved a la RTS games.

- Common elements: **Health bars** (for your character or units under your control), **ability buttons**, and a **minimap**.

- Aesthetically, UI elements have a **dark fantasy motif** – ornate frames, Gothic font for headings, but high readability. We avoid tiny text. Important elements have distinct colors (heroes' health bars green, Dark Lord units' health bars red, etc.).

Hero UI & Controls:

Each hero is controlled similarly to a MOBA hero or action RPG character for familiarity:

- **Movement:** Right click with mouse or WSAD keys. Smooth movement is important for kiting monsters. Heroes can also attack-move: if an enemy is nearby, the hero auto-attacks while moving (if in range).

- **Abilities:** Buttons on the right side for abilities:

    - Typically 3 ability buttons + 1 ultimate (if applicable) + perhaps a potion/consumable quick slot. For example, the Knight would have buttons for Shield Bash, Charge, Defensive Stance, and an ultimate like Whirlwind. Each shows a cooldown overlay when used.

    - Targeting: Some abilities might require targeting a location or enemy (e.g. Witch's curse – click button then click target monster or ground). On mobile, we allow dragging an ability button onto the map to choose direction or target (a common practice in mobile MOBAs like *Wild Rift*). Simpler abilities auto-target nearest foe or self if it's a buff, to reduce mis-clicks.

- **Basic Attack:** Either a dedicated attack button (for ranged heroes like Witch if she has a basic attack spell) or just clickping an enemy. We might include a button to toggle

auto-attack on/off in case players want to stutter-step manually.

- **Hero HUD:** Shows the hero's portrait, HP bar, and resource (mana or stamina if they use one – e.g. Witch's mana, Knight perhaps doesn't have mana but a stamina for special moves). Also shows any active buffs/debuffs (small icons). For example, if Alchemist drank a strength potion, an icon appears with its remaining duration.

- **Hero Special Resource UI:** As mentioned, unique resources: Goblin's gold count displayed as a gold coin icon with number, Alchemist's ingredient bag (possibly clickping it opens a quick crafting menu where she can drag two ingredients to combine – this menu pauses her movement or she must be stationary and safe to use it; doing it in real-time might be too hard while moving). The Knight's level and XP bar shown near his portrait (like "Level 2 – XP: 40/100"). The Witch's shrine status perhaps as small icons top of screen – or integrated into minimap (shrines on map glow green or red if cleansed or not).

- **Minimap & Team Info:** Heroes have a minimap top-right (for example) that shows terrain and pings. Fog of war is not as relevant to them, but they won't see monster icons unless spotted by them. It will show ally positions as colored dots (each hero has a color or icon). clickping the minimap could enlarge it for more detail (but opaque enough to still see if moving). Team status UI might list allies' HP (a small bar for each teammate on side of screen) so you know if a friend is in trouble even off-screen. Possibly an icon will flash if a teammate is downed.

- **Interaction Prompts:** When a hero is near something interactive (villager, chest, shrine), a button prompt appears (e.g. "click to interact" or a hand icon). Since screen space is limited, we do context-sensitive: the same button might serve as interact when near an object, otherwise not shown.

- **Communication:** A chat or ping button opens a radial menu with quick phrases ("Help!", "Enemy here!", "Gathering ingredients", etc.). Also possibly voice-overs for these for flavor (Knight shouts "To me, allies!" when "Help!" pinged, etc.).

- **Feedback:** Screen shake and hit flashes for combat hits (subtle on mobile). Damage numbers floating for clarity (optional toggle). When a hero dies, their UI could gray out and a death summary shown ("Killed by Ogre – lasted 7m"). They can then spectate allies (cycle through views).

Unit Controls:

The Dark Hero's interface is more akin to an RTS god-view:

- **Unit Management:** The right side might list current units (portraits or symbols, with health bars). The Dark Hero can click a unit's icon to select it (camera jumps to it, and it gets highlighted). Selecting multiple units might be tricky; instead, we can use group commands: e.g. a "Select All" button or select by type (click and hold Scout icon could select all scouts). But we expect at most ~10 units, so clickping each is feasible. Once a unit is selected, the Dark Hero can issue commands:

    - **Move/Patrol:** click on map or minimap to set a destination/patrol point. A radial might ask "Move here" vs "Patrol this area" difference (patrol could mean the unit will then roam around that point).

    - **Attack Target:** If an enemy hero or neutral is spotted and visible on screen, clickping it while a unit is selected orders an attack focus. If Fog, you can't target directly until visible.

    - **Abilities:** Most units have passive AI for abilities, but if any special actions (like Priest's reveal spell) can be manually triggered, we'd have those as small buttons when unit selected. We may decide to not allow manual cast to keep Dark Lord focus on macro, letting AI handle it – probably better for simplicity.

    - Possibly a **Rally All** button to call all units back to castle if needed (panic button if heroes are at gates).

- **Castle UI:** The castle itself can be selected. Doing so shows options like upgrading a tower, locking a gate (if that's a mechanic), or using a **Castle Ability**. For example, if the Castle has a "Reveal Map" spell, that might be a button available when castle is selected (or always on UI if it's global). We might have a dedicated ability bar for the Castle for its spells like the heroes do: e.g. bottom right has Scrying Orb and Fear Surge icons with cooldown. He's kind of like a hero too in final fight, so he might have a basic attack or spell if directly clicked. But since he's stationary until late, those abilities are more strategic.

- **Alerts & Notifications (Dark Hero):** If a unit spots a hero or hears something, an **alert ping** appears on the UI (e.g. the edge of screen flashes in the direction, and the minimap pings red). clickping the alert could snap camera there. Also, if a unit is attacked or killed, similar alerts (e.g. "Scout #2 slain"). Possibly an audio cue like the Dark Hero's advisor (if he has a minion servant voice) says "Our scout has perished, my Lord." The Dark Hero also sees notifications of major hero actions that he could plausibly notice: shrines cleansed (as mentioned), tavern found (if one of his units stumbles on the tavern location, an icon marks it thereafter).

- **Team Info:** A UI for his units' status (the list of units already covers HP, maybe an icon if they are in combat). Also possibly a **mini-log** that says "Shrine cleansed" or "Village A

raided" for events.

- **Communication:** TBD

Menus and Out-of-Game UI:

Polished menus for character selection, matchmaking, etc.:

- **Main Menu:** Dark fairytale themed background (maybe the Dark Lord's castle silhouette). Options: Play as Dark Lord, Play as Heroes (which then either assigns a random hero or lets you pick which hero if available), Tutorials, etc.

- **Hero Select:** If playing in a party or custom game, players can choose which hero they want (ensuring 4 different picks). The Dark Lord slot is unique. In quick match, perhaps you queue separately as Dark Lord or Hero. The UI will show if one side has longer queue times (likely more want to play hero than villain or vice versa, we'll adjust incentives accordingly).

- **Tutorial & Codex:** There will be an in-game codex explaining heroes, monsters, and map elements, accessible from menu. It will have images and short text so players can learn (since it's a complex game, we want an easy reference).

- **HUD customization:** Some players might want to move buttons or change size – if feasible we'll allow some customization of control layout, as many mobile games do for comfort.

Art Integration in UI:

The UI art direction aligns with the cartoony-gothic style. For instance, the hero portraits might be drawn like painted fairytale characters with exaggerated features. The ability icons use a storybook-like iconography. The health and mana bars could look like vials of potion or glowing crystal shards. We will ensure that while thematic, the UI maintains clarity (e.g. not too ornate to read numbers).

Accessibility:

Consider colorblind options (distinct shapes on minimap, not just colors). Large text mode for small screens. Perhaps haptic feedback on important alerts (phone vibrates when your hero is low HP or when Dark Lord's unit spots a hero). These help players who might otherwise miss subtle cues.

In summary, the UI is designed to make controlling a single hero intuitive even on a phone, and to make controlling an army as the Dark Lord possible without a mouse/keyboard. Simplicity where possible: click, drag, and a few buttons. Testing will refine touch targets and responsiveness. We want players to feel **in control** of their character or army without fighting the interface – any frustration should come from the intense gameplay, not from mis-clicking an ability.

Art Direction

The art and visual design of *Grimm Dominion* is crucial to establishing its dark fairytale identity and ensuring the tone balances creepy and comedic. This section covers visual style, characters, environments, and audio-visual tone (though audio is not heavily detailed, we mention it for completeness).

Visual Style Overview:

We are aiming for a **"cartoony but moody"** aesthetic – similar to how *Warcraft III* achieved a colorful yet gothic look (Reforged graphics, art direction - General Discussion - Warcraft III: Reforged Forums). Characters and creatures will have exaggerated proportions and expressive animations (cartoon influence), making them readable on small screens and giving a slightly humorous vibe. At the same time, the color palette and lighting lean dark, with lots of shadows, rich deep colors (purples, dark greens, blacks), and highlights of glowing magic or fire to create contrast. Think of a twisted fairy tale illustration: whimsical shapes with a sinister twist.

- **Inspirations:** Classic Blizzard art from Warcraft III and *The Frozen Throne*, Tim Burton's *Nightmare Before Christmas* (for gothic whimsy), and games like *Overlord (2007)* which used dark humour and satirical fantasy art (Overlord (2007 video game) - Wikipedia). Also, illustrations from Grimm's fairy tales where forests are dark and characters stylized.

- **Characters:** Each hero and unit is designed with a strong silhouette and exaggerated features to be identifiable in game view. For example:

    - *Goblin:* Small hunched figure with oversized ears and a big sack of gold; sneaks with comical tip-toe animations. Colors: earthy greens and browns to blend in, with maybe a red bandana for a splash (and cosmetic potential).

    - *Alchemist:* Perhaps a short and stout character with a huge backpack laboratory. Vials and potions visibly clinking. Animations: mixing potions frantically, with cartoonish explosions (so even a failed craft is visually funny).

    - *Knight:* Broad-shouldered, shining (but scuffed) armor, maybe a ludicrously large sword to emphasize his archetype. Gestures boldly. Even his idle stance is heroic

(hand on hip, chest out).

    - *Witch:* Flowing ragged cloak that moves mysteriously, possibly slight glow or particle effects around her to hint at magic. Could have a companion like a crow on her shoulder (that also can serve as an animation element, like sending it out for scouting maybe).

    - *Dark Lord:* We need a visually compelling villain. Perhaps tall, slender, clad in ornate armor with spikes, or a shadowy phantom king. Since he's stuck in a castle, maybe he's often depicted on a throne or balcony. Expressive animations when giving orders (waving a hand to summon a unit, etc.). He should be intimidating but also a bit comical in arrogance – e.g. when heroes breach his throne room, he might do a dramatic facepalm or exaggerated rage animation.

    - *Monsters:* Each monster unit type looks distinct:

        - Scouts could be imp-like or small winged gargoyles, with huge eyes (for seeing far) but diminutive bodies.

        - Tanks like ogres or golems with massive upper bodies, tiny legs (for comedic effect on slowness).

        - Priests as gaunt robed figures with oversized heads or hats, maybe glowing staffs; they look frail and pompous.

        - Keep consistency that monster designs hint at their behavior: e.g. a Priest might constantly be looking down at a book (ignoring surroundings unless needed).

- **Animation & Humor:** Animations will inject humor – Goblin rubbing his hands when he gets gold, Knight striking a pose after leveling up, Dark Lord's minions scratching their heads when they lose sight of a hero. Death animations can be over-the-top (monsters explode into bones, heroes get knocked into the air). We will also incorporate *emotes* that characters do contextually or that players can trigger (cosmetic emotes monetize later) – e.g. Dark Lord slow-clapping sarcastically if he kills a hero, the heroes dancing or teasing when they evade enemies.

- **Environments:** The kingdom is a patchwork of fairytale locales but twisted:

    - The **Dark Lord's castle** is gothic and looming – high contrast of sharp silhouettes (spires, gargoyles). It's central and visible from afar as a landmark. We might show the castle in the backdrop skybox from any point (to remind players of the ultimate goal). Interiors of the castle (for final fight) have grand halls with stained glass (maybe depicting comedic versions of the Dark Lord's

"glorious" deeds). Lighting in castle is eerie (torches casting long shadows, flicker).

- **Forest biome:** Gnarled trees with faces in bark, thick underbrush. Possibly signs of fairytale creatures (like a cottage that might recall Red Riding Hood's but abandoned). Lighting: moonbeams through branches.

- **Swamp biome:** Murky, with glowing will-o-wisp lights. Bubbling green water. Good place for Alchemist ingredients (colored mushrooms, etc.). Perhaps the ruins of an old church sunk in mud (a shrine location).

- **Village biome:** Cozy cottages but currently fearful. We can make them cute but also show distress (boarded windows or hiding villagers). When monsters come, villagers run and scream (cartoonishly). If Knight saves them, their mood brightens (lighting literally could change a bit warmer in that village).

- **Graveyard biome:** Crooked tombstones, mist, maybe undead neutrals. Witch shrine likely here (with bluish ghostly light).

- **Mountains/Caves:** Rocky areas with narrow paths, maybe a dragon's skeleton as a landmark. Dark Lord might have a scout or shrine here guarded by something.

- **Overall palette:** Each area has its dominant colors but all under a "nighttime" filter if match is at night (we might set matches at eternal dusk for mood). Colors pop for interactive elements (e.g. gold glints, ingredient herbs have slight glow). Heroes each have a thematic color too (Goblin green/gold, Alchemist purple, Knight blue/silver, Witch dark green/purple) to help identify them in the environment.

- **UI Art:** As mentioned, UI frames are ornate. E.g., the hero portraits could be in a little mirror or storybook icon. The menus might have a backdrop that looks like an open magical tome with moving illustrations. Loading screens can show storybook-style art of the characters with witty captions ("Tip: Goblins don't make good friends, but they make great thieves!").

Tone and Narrative Flavor:

We aren't writing a full narrative here, but the setting is communicated through visuals and a bit of writing.

- The **dark fairytale tone** means we can reference known fairytales but subvert them. Perhaps one of the heroes (not explicitly listed) could be a twisted fairytale character

(e.g. a Wolf-Man that was once the Big Bad Wolf but now fights the greater evil). This could appear as future DLC hero or in lore.

- The humor is **wry and satirical**: The Dark Lord is a bit genre-savvy ("Why do they always hide in the woods? Ugh." he might mutter). Heroes banter: the Knight's earnestness vs Goblin's greed can cause snappy jokes. We ensure the humor is light and doesn't undermine the stakes entirely – it's similar to *Overlord (2007)*'s satire of fantasy (Overlord (2007 video game) - Wikipedia), or *Shrek* in a darker form.

- For example, if a hero dies, maybe the Dark Lord comments "Another one bites the dust… how disappointing, I expected more." If the Dark Lord loses a unit, maybe he yells "Imbeciles! I have to do everything myself?"

- These lines would be delivered in voice acting ideally, adding personality (with an option to mute for those who prefer silent). The voice acting style could be hammy on purpose (like an exaggerated villain voice, a sly goblin voice, etc.).

Audio & Music: (briefly)

- **Music:** A dynamic score that mixes whimsical and dark. Possibly a fantasy orchestral base with some playful instruments (like a harpsichord for the Dark Lord's castle scenes, playful flutes in the forest that turn ominous when enemies appear). During tense chases, the music ramps up tempo. In stealthy moments, it tones down to ambient drones. We could even have different leitmotifs for each hero and the Dark Lord, interweaving as they clash.

- **Sound Effects:** Cartoony exaggeration but with a dark edge. E.g. when the Goblin picks up gold, a classic coin jingle plays. When the Witch casts a curse, a creepy whisper sound. Monster growls are somewhat comical (an ogre's stomach rumbles loudly, a scout imp chatters annoyingly). These give audio feedback and charm. UI sounds like clicking menu might sound like flipping a page or clinking a potion.

- **Voices:** As mentioned, characters with one-liners. We will be careful to not make them too repetitive or irritating; they should spice the experience. Possibly allow heroes to have a few emote lines players can trigger (some free, some premium for fun).

Cosmetics and Variation:

Even though gameplay monetization is cosmetics-only, we design from start to allow many skins and visual variants:

- Each hero can have alternate costumes that still fit their silhouette (so gameplay clarity remains). E.g. a “Royal Knight” skin with shiny polished armor and a plume, or a “Witch of the West” skin for the Witch with different color scheme and hat. These skins maintain the general vibe but allow personal expression.

- The Dark Lord could have skins referencing famous villains (parodies), like a vampire count look, or a evil clown (for a whimsical event).

- Monster skins: maybe you can turn all your summoned units into gingerbread cookie versions for a holiday (humor: gingerbread men attacking heroes).

- VFX changes: Cosmetic spells – a potion could explode into fireworks instead of smoke, etc. These will be careful to not confuse players (they’ll know it’s cosmetic via consistent effects).

- Environment cosmetics: not as likely, but maybe the Dark Lord can choose a theme for his castle that other players see (like a Halloween theme with pumpkins, purely visual). Could be fun seasonal content.

Performance considerations for art: We target mobile, so we will use stylized textures (hand-painted look, which is less heavy than realistic high-res). Polycounts for models will be moderate, using normal maps to give detail without heavy geometry. We’ll also allow lowering detail in settings (simplify shadows, reduce particle effects) for lower-end devices. The cartoon style lends itself to being more forgiving in lower fidelity. Characters will be maybe around 5-10k polygons each at most, given 5 players plus ~10 monsters could be on screen at times. We will use level-of-detail (LOD) models for monsters when far away or many on screen.

In essence, the art direction aims to make *Grimm Dominion* immediately visually appealing and thematically coherent: a world that feels like a storybook come to life, albeit a storybook where the Big Bad Wolf might slip on a banana peel while chasing Little Red Riding Hood. The combination of dark atmospheric design with humorous execution will set it apart and align with the gameplay’s mix of tension and fun.

Technical Considerations

Delivering this game on mobile with real-time 1v4 multiplayer, AI, and procedural content requires careful technical planning. Here we outline the key technical needs and how we plan to address them:

Platform & Engine:

- We will use a robust cross-platform engine such as **Unity** or **Unreal Engine (UE4/UE5)**, both of which have strong mobile support and tools for networking, AI, and rendering.

Unity is known for mobile optimization and many mobile multiplayer games use it, so it's a likely choice. Unreal offers great rendering but might be heavier on low-end devices; we will evaluate, but assume Unity for now.

- Target platforms: iOS (iPhone 8 and above, iPad 5th gen and above) and Android (mid to high tier devices from last ~4 years). We target at least 30 FPS performance on recommended devices, with possible 60 FPS on high-end.

- The game engine will support both **touch input** and potentially controllers (for those who connect a Bluetooth controller to their device, though UI will be built around touch primarily).

Networking & Multiplayer Architecture:

- A 4v4 multiplayer game requires a network architecture to handle 8 concurrent players. We plan to implement an **authoritative server model**: a central server (in the cloud) simulates the game state to prevent cheating (especially important since one player (Dark Lord) having map hack or such would ruin balance). Players' devices are clients sending inputs.

- We'll use a service or library for real-time multiplayer (e.g. Photon, UNet, or a custom dedicated server). The authoritative server also needs to handle the **AI behaviors** consistently (so AI isn't desynced between players).

- We will optimize network messages by focusing on player commands and significant events rather than every small movement (predictive movement can be done client-side for smoothness). Given the RTS aspect, the Dark Lord might issue fewer commands than a typical action game, which helps.

- **Latency considerations:** We aim for region-based servers to keep ping low (under 100ms ideally). The game's design somewhat tolerates mild latency because unit AI has inherent delays (not all twitch reactions). However, dodging a monster or using an ability still needs responsiveness, so we'll use techniques like client-side prediction for hero movement and server reconciliation.

- **Matchmaking:** We will have a system to queue players. Players can queue as Dark Lord or Hero. If too many queue as one, we might incentivize the other (like bonus progression if you play the less popular side). MMR (matchmaking rating) or casual modes will be considered – likely an ELO-like rating to match experienced players together for balance. A group of four can queue as heroes together, then the system finds a Dark Lord of similar skill. Cross-platform play iOS/Android is supported since same engine.

AI Implementation:

- AI for monsters will be implemented via behavior trees or state machines in the engine. Unity has tools or we might code our own. The AI must run on the server to be authoritative, but client can run a predictive copy for immediate feedback (especially Dark Lord's client, to see his unit respond without waiting for round trip).

- Pathfinding: We will use a navigation mesh (NavMesh) for the map that AI agents use to move around obstacles. Unity's NavMesh can handle dynamic obstacles (we'll mark moving heroes as dynamic obstacles so monsters can chase properly). We need to ensure performance with possibly up to 10 AI units pathfinding simultaneously – which is doable on server side given modern CPU, but for safety we limit very large hordes.

- Procedural generation on server: The server will generate the map layout at match start (using a seed) and then send the relevant map data to clients (or clients generate it from the seed to save bandwidth). This includes where walls, obstacles, interactive objects are so that the nav mesh can be built accordingly on both server and client.

- AI difficulty scaling: We may implement different AI profiles (for a possible PvE mode or difficulty settings), but in PvP the AI is essentially the Dark Lord's tool, so difficulty is indirectly controlled by how many/which units he spawns.

Procedural Generation Tech:

- The map will be constructed from pre-made **tiles or chunks**. For example, we design a set of forest clearing patterns, village layouts, etc. The procedural generator then picks and stitches these. This ensures we don't generate completely random geometry that could glitch AI; instead, each chunk is tested to work. We might use a grid or graph-based placement.

- The castle is a fixed chunk at center. Then rings of chunks around it maybe.

- We will need a random seed per match to allow debugging: if a specific map arrangement causes imbalance issues, we can reproduce it.

- The generator runs on server (ensuring consistency) and transmits at least the high-level results to players (like locations of key points). The client might then instantiate the appropriate tiles. We must ensure the random generation is reasonably fast (a couple seconds) to not delay match start too long.

- Memory: We'll reuse textures and models across these chunks heavily (like modular walls, tree models reused). Procedural means a bit more load as more unique layout, but

we keep it optimized by instantiating from a pool of assets.

Performance & Optimization:

- **Level of Detail (LOD):** As mentioned, use simpler models or even billboards for units far from the camera. On mobile, the screen isn't huge, so we can aggressively swap to low poly or 2D impostors at distance.

- **Culling:** Frustum culling and possibly fog/hide units not in view to reduce draw calls. Heroes have limited view, but Dark Lord can see multiple areas via units; still, we won't render what's not on screen.

- **Particle Effects:** Optimize for mobile – use CPU particles sparingly or GPU particles if supported. Too many potion sparkles or magic glows can tank performance, so we will limit concurrent effects. Perhaps degrade effect quality on older devices (less particles, simpler shaders).

- **Physics:** Minimal. We might use physics for ragdoll deaths or thrown objects like physics-based grenades, but keep it lightweight (predefined animations might suffice for many). No heavy physics simulation continuously (no need for complex cloth sim etc., though Unity's lightweight cloth could add cape movement cheaply).

- **Memory:** Aim to fit comfortably in under ~500 MB RAM usage on device. We'll have to manage asset sizes (textures compressed, sound streaming when possible). Unity's addressable assets or asset bundles can help load only needed chunks.

- **Battery and Thermal:** Mobile gaming can heat devices; we allow frame rate cap or quality settings. We test on device for thermal throttling. Possibly provide an option for 30fps lock to save battery if players prefer.

Testing & Tools:

- We will create a **spectator/replay tool** internally to watch matches to observe balance (especially important in asymmetrical games to see if one side consistently dominates).

- **Analytics:** Track metrics like average match time, hero win rate vs Dark Lord, usage of units, etc., to adjust balancing.

- **Cheat prevention:** Because it's competitive, in addition to authoritative server, we'll watch for anomalies (like a hero moving impossibly fast – could indicate a sped-up client hack; server would override anyway). Also hide critical info on client (heroes' positions

obviously not sent to Dark Lord's client until needed).

- **Online requirements:** Likely always-online due to PvP focus. A fallback single-player vs AI mode could be considered for practice, which would run everything locally, but core is online.

Compatibility and Future-Proofing:

- We plan to support updates with new content, so an asset delivery system is needed (so users don't redownload entire app for every update, or using app stores effectively).
- Cross-play between iOS and Android is a goal, as mentioned.
- Possibly consider a PC version later (maybe via an emulator or separate build) if demand, but initially focus on mobile.

In summary, technically this project is ambitious but feasible: similar scope to a mobile MOBA or RTS. We leverage existing engine tech for most heavy lifting (Unity's netcode or Photon for networking, NavMesh for AI pathfinding, etc.). The key is optimizing for 5 players + AI on mobile hardware and ensuring a smooth user experience. With careful planning and iterative testing, we can achieve stable performance. It's crucial to hit that sweet spot where the game looks good and runs well, because any lag or frame drop could frustrate players in a tight match.

Monetization Strategy

Grimm Dominion will follow a **"cosmetics-only" monetization model**, meaning we will not sell any gameplay advantages – all heroes, abilities, and power progression are unlocked through gameplay. This fair approach is important for a competitive game and is proven viable by successful games like League of Legends and Dota 2 (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar) (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar). Our monetization will focus on selling cosmetic items and convenience features that do not affect match outcomes. Below we detail the monetization and how it integrates with player progression.

Free-to-Play Model:

The game will be free to download and play. All core content – the Dark Lord role, the four starting heroes, all maps (since procedural), and future gameplay additions – are accessible to everyone. Revenue comes from players purchasing cosmetic customizations, seasonal battle passes, or optional out-of-match content.

Cosmetic Items:

We will offer a wide array of cosmetic items for those who want to personalize their experience:

- **Character Skins:** New outfits or models for Heroes and the Dark Lord. For instance, a Goblin skin where he's dressed as a highwayman with a tiny cape, or a Dark Lord skin that turns him into an undead lich appearance. Skins can be premium (paid) or earned through events. They change character appearance, and may have unique idle animations or ability VFX (without altering ability function). For example, a Witch skin might change her spell colors to blue instead of green – always mindful of clarity (we won't confuse fire vs ice damage, it's just visual flair).

- **Unit Skins:** The Dark Lord can buy skins that change how his minions look. E.g., a "Toy Army" skin pack makes Tanks look like giant wind-up tin soldiers and Scouts like toy planes, for a humorous effect. These are purely visual; to other players it's obvious they're just reskinned (we ensure silhouette or an indicator remains clear).

- **Emotes & Animations:** Emotes are brief animations/voice lines a player can trigger (like a dance, taunt, cheer). For heroes, these can be used during downtime or after a victory. For the Dark Lord, emotes might play in a little picture-in-picture of him reacting (because his actual avatar might not always be visible to heroes). This adds fun and player expression. Cosmetic emotes could be sold or earned.

- **VFX and Audio Packs:** Possibly allow customizing certain effects. For example, a cosmetic that changes the Dark Lord's global announcement voice (imagine a silly high-pitched narrator voice as a joke option during April Fool's event). Or different footstep effect colors for heroes. These are minor and need to be careful not to annoy other players; mostly client-side or subtle differences.

- **Profile Cosmetics:** Avatars, profile banners, special icons for number of wins, etc. These show off in lobbies or matchmaking screens. Not gameplay, but bragging rights.

Quality and Pricing: Cosmetics will range in complexity (common recolors vs. legendary full model changes) and be priced accordingly. We'll use a soft currency + premium currency model:

- Players earn a little soft currency by playing (e.g. gold coins) which can unlock some cheaper cosmetics over time or be saved to partially discount a purchase.

- A premium currency (gems) bought with real money can unlock cosmetics directly. We'll ensure players can also get premium currency via battle pass or events in small amounts (free track), but major purchases likely need either lots of grinding or a purchase.

- Prices might range from $2 for a simple recolor skin to $10-15 for elaborate skin packs. Emotes maybe $1-3. We'll adjust to market norms in mobile F2P.

Importantly, as noted, these cosmetics **do not confer any competitive advantage**, merely personal satisfaction and status. This fosters goodwill in the community as it's seen as fair monetization (Is Cosmetic only monetization viable in the long run? - Reddit). Players who spend support the game's development, but those who don't spend can still enjoy full gameplay.

Battle Pass (Seasonal):

We plan to implement a **seasonal Battle Pass** system, which is common in F2P games for steady monetization. Each season (say every 2-3 months, possibly aligned with a theme or new hero/unit release) we have:

- A free track: all players progress by playing matches, unlocking some rewards (currency, a few cosmetics, maybe a loot box).

- A premium track: costs around $10 and provides additional rewards at each tier (including exclusive cosmetics, larger currency gains, unique seasonal skin at the end). If a player completes the pass, they might earn enough premium currency to buy the next one (incentivizing continued engagement) (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar).

- The pass content ties into the theme – e.g. a "Frozen Throne" season might have icy-themed skins for each hero and Dark Lord in the pass.

- Challenges/quests to accelerate pass progression (like "Use the Goblin's backstab 20 times" or "Win 5 matches as Dark Lord").

- The battle pass also serves to drip-feed content and encourage players to try roles they might not (if challenges push them to).

Meta-Progression & Monetization:

All meta-progression (unlocking abilities, heroes, etc.) is done via gameplay, not pay. We do **not** sell XP boosters for hero progression (to avoid perception of pay-to-win). The only potential "convenience" purchase might be something like a doubled cosmetics-earning rate or extra daily challenges, but again we tread carefully. If any convenience item touches progression, it must not unbalance – likely we skip it or limit it to cosmetic progression only. We may sell additional **loadout slots** or something trivial if needed, but currently seems unnecessary.

Ads and Currency:

We do not plan to use intrusive ads. Possibly optional rewarded ads (e.g. "watch an ad for a small amount of soft currency or a loot box key") for truly free players to get some cosmetics

without purchase. But we have to keep the gothic tone; ads can break immersion, so this might be only in a main menu context, not in-game obviously.

Community and Cosmetic Show-off:

Cosmetics monetization thrives when players can show off (The Cosmetic Monetization Trend: How to Sell Cosmetic Items?). We ensure that in pre-game lobbies or at match end MVP screens, players' skins and emotes are visible to others. For example, if a hero team wins, the end screen shows the four heroes posing with their skins and perhaps a title ("Goblin – Master Thief") under the one with highest gold stolen, etc. The Dark Lord if wins gets a screen showing him triumphant on his throne (with his skin). This encourages envy/aspiration ("Cool skin, I want that too.").

Additionally, we'll run seasonal events (e.g. Halloween event with free pumpkin head cosmetic for all who play that week, Christmas with a Santa goblin skin, etc.) to keep the cosmetic offerings fresh and players checking in. Even though these are "free" cosmetics, they engage users who might then purchase others.

No Pay-to-Win Guarantee:

We will be transparent that spending money will **never** give gameplay edge. This fosters a trust with players – very important in competitive communities. As one Redditor succinctly noted, games can be extremely successful focusing on cosmetic-only models because players appreciate fairness (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar). We intend to uphold that.

For example, if we ever introduce a new hero or monster type as DLC, it would be a free update or unlocked by playing, not sold for money. (If down the line monetization needed a boost, maybe new heroes could be unlocked with grind or bought to skip grind, similar to LoL champions, but since we said cosmetics-only, we'd avoid even that unless absolutely acceptable to community).

Cosmetic Creation & User Content:

Over time, as community builds, we could involve them in design contests for skins (fairytale theme is ripe for fan ideas). This increases buy-in when the skin releases (people want to buy the skin they or their peers designed).

Finally, we'll monitor monetization metrics ethically: ensuring we provide value and don't push too hard. The game's longevity will come from a passionate player base that feels their purchases are meaningful cosmetics and that the game isn't gouging them. This aligns with the established trend that fair **cosmetic monetization can drive success when paired with a strong gameplay loop** (Is Cosmetic only monetization viable in the long run? - Reddit).

Sources: The design above draws inspiration from asymmetric multiplayer precedents and fantasy themes. For instance, Warcraft III's *Island Defense* mod established the one-vs-many strategy concept (Island Defense v3.0.3g | HIVE), and popular 4v1 games like *Dead by Daylight* have shown the lasting appeal of such formats (How Asymmetrical Multiplayer Flop 'Evolve' Paved the Way For 'Dead By Daylight' and 'Friday the 13th' - Bloody Disgusting). The dark humor tone references games like *Overlord* which parodied fantasy tropes (Overlord (2007 video game) - Wikipedia). Our fair monetization philosophy is informed by the success of cosmetic-only models in top games (Is Cosmetic only monetization viable in the long run? : r/PlayTheBazaar). By combining these insights with original mechanics suited for mobile, *Grimm Dominion* aims to deliver a fresh, engaging, and enduring experience.